

- 1 (a) (i) *either* rate of change of displacement  
or (change in) displacement / time (taken) B1 [1]
- (ii) speed has magnitude only B1  
velocity has magnitude and direction B1 [2]
- (b) (i) idea of area under graph / use of  $s = \frac{(u + v)}{2} \times t$  C1
- $s = \frac{(18 + 32)}{2} \times 2.5$  C1  
= 62.5 m A1 [3]
- (ii)  $a = (18 - 32) / 2.5 (= -5.6)$  C1  
 $F = ma$  C1  
 $F = 1500 \times (-) 5.6 = (-) 8400 \text{ N}$  A1 [3]
- (c) arrow labelled A and arrow labelled F both to the left B1 [1]